

Vehicle Registration System

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1) main.cpp

```
#include <iostream>
#include "vehicle.h"
using namespace std;

int main() {
    int choice;
    do {
        cout << "
=== Vehicle Registration System ===
";
        cout << "1. Add new record
";
        cout << "2. Display all records
";
        cout << "3. Search by plate number
";
        cout << "4. Exit
";
        cout << "Enter choice: ";
        cin >> choice;

        if (choice == 1) addRecord();
        else if (choice == 2) displayAll();
        else if (choice == 3) searchRecord();
        else if (choice == 4) cout << "Goodbye!
";
        else cout << "Invalid choice. Try again.
";
    } while (choice != 4);

    return 0;
}
```

2) vehicle.cpp

```
#include "vehicle.h"
#include <iostream>
#include <fstream>
#include <sstream>
#include <iomanip>
using namespace std;
```

```

User::User() {}

User::User(string n, string e, string p) {
    name = n;
    email = e;
    phone = p;
}

string User::getName() { return name; }
string User::getEmail() { return email; }
string User::getPhone() { return phone; }

void User::setName(string n) { name = n; }
void User::setEmail(string e) { email = e; }
void User::setPhone(string p) { phone = p; }

VehicleOwner::VehicleOwner() {}
VehicleOwner::VehicleOwner(string n, string e, string p) : User(n, e, p) {}

Vehicle::Vehicle() {}
Vehicle::Vehicle(string plate, string type, int yr, VehicleOwner o) {
    plateNumber = plate;
    vehicleType = type;
    year = yr;
    owner = o;
}

string Vehicle::getPlate() { return plateNumber; }
string Vehicle::getType() { return vehicleType; }
int Vehicle::getYear() { return year; }
VehicleOwner Vehicle::getOwner() { return owner; }

bool isValidEmail(string email) {
    int atPos = email.find('@');
    int dotPos = email.rfind('.');
    if (atPos == string::npos || dotPos == string::npos) return false;
    if (atPos == 0 || dotPos < atPos) return false;
    if (dotPos == email.length() - 1) return false;
    return true;
}

bool isValidPhone(string phone) {
    if (phone.length() != 13) return false;
    if (phone.substr(0, 4) != "+250") return false;
    for (int i = 4; i < phone.length(); i++) {
        if (!isdigit(phone[i])) return false;
    }
    return true;
}

```

```

bool isValidPlate(string plate) {
    if (plate.length() != 7) return false;
    if (!isalpha(plate[0]) || !isalpha(plate[1]) || !isalpha(plate[2])) return false;
    if (!isdigit(plate[3]) || !isdigit(plate[4]) || !isdigit(plate[5])) return false;
    if (!isalpha(plate[6])) return false;
    return true;
}

void saveRecord(Vehicle v) {
    ofstream file("records.txt", ios::app);
    file << v.getPlate() << ", "
        << v.getType() << ", "
        << v.getYear() << ", "
        << v.getOwner().getName() << ", "
        << v.getOwner().getEmail() << ", "
        << v.getOwner().getPhone() << "
    ";
    file.close();
}

vector<Vehicle> loadRecords() {
    vector<Vehicle> records;
    ifstream file("records.txt");
    string line;
    while (getline(file, line)) {
        stringstream ss(line);
        string plate, type, yearStr, name, email, phone;
        getline(ss, plate, ',');
        getline(ss, type, ',');
        getline(ss, yearStr, ',');
        getline(ss, name, ',');
        getline(ss, email, ',');
        getline(ss, phone, ',');

        int year = stoi(yearStr);
        VehicleOwner owner(name, email, phone);
        Vehicle v(plate, type, year, owner);
        records.push_back(v);
    }
    file.close();
    return records;
}

void addRecord() {
    string plate, type, name, email, phone;
    int year;
    do {
        cout << "Enter plate number (e.g. RAA123A): ";
        cin >> plate;
        if (!isValidPlate(plate))
            cout << "Invalid plate. Must be like RAA123A."

```

```

";
    } while (!isValidPlate(plate));

    cout << "Enter vehicle type (e.g. Sedan, SUV): ";
    cin >> type;
    cout << "Enter year of manufacture: ";
    cin >> year;
    cin.ignore();
    cout << "Enter owner name: ";
    getline(cin, name);

    do {
        cout << "Enter owner email: ";
        cin >> email;
        if (!isValidEmail(email))
            cout << "Invalid email. Must contain @ and .
";
    } while (!isValidEmail(email));

    do {
        cout << "Enter phone number (e.g. +250788123456): ";
        cin >> phone;
        if (!isValidPhone(phone))
            cout << "Invalid phone. Must start with +250 and be 13 characters.
";
    } while (!isValidPhone(phone));

    VehicleOwner owner(name, email, phone);
    Vehicle v(plate, type, year, owner);
    saveRecord(v);
    cout << "Record saved successfully!
";
}

void displayAll() {
    vector<Vehicle> records = loadRecords();
    if (records.empty()) {
        cout << "
No records found.
";
        return;
    }
    cout << "
";
    cout << left
        << setw(10) << "Plate"
        << setw(10) << "Type"
        << setw(7) << "Year"
        << setw(18) << "Owner"
        << setw(25) << "Email"
        << setw(15) << "Phone"

```

```

        << "
";
    cout << string(85, '-') << "
";
    for (int i = 0; i < records.size(); i++) {
        Vehicle v = records[i];
        cout << left
            << setw(10) << v.getPlate()
            << setw(10) << v.getType()
            << setw(7) << v.getYear()
            << setw(18) << v.getOwner().getName()
            << setw(25) << v.getOwner().getEmail()
            << setw(15) << v.getOwner().getPhone()
            << "
";
    }
}

void searchRecord() {
    string plate;
    cout << "Enter plate number to search: ";
    cin >> plate;
    vector<Vehicle> records = loadRecords();
    bool found = false;
    for (int i = 0; i < records.size(); i++) {
        if (records[i].getPlate() == plate) {
            cout << "
- Vehicle Found -
";
            cout << "Plate Number: " << records[i].getPlate() << "
";
            cout << "Vehicle Type: " << records[i].getType() << "
";
            cout << "Year          : " << records[i].getYear() << "
";
            cout << "Owner Name   : " << records[i].getOwner().getName() << "
";
            cout << "Email        : " << records[i].getOwner().getEmail() << "
";
            cout << "Phone         : " << records[i].getOwner().getPhone() << "
";
            found = true;
            break;
        }
    }
    if (!found) {
        cout << "Vehicle with plate " << plate << " not found.
";
    }
}

```

3) Vehicle.h

```
#ifndef VEHICLE_H
#define VEHICLE_H

#include <string>
#include <vector>
using namespace std;

class User {
private:
    string name;
    string email;
    string phone;
public:
    User();
    User(string n, string e, string p);
    string getName();
    string getEmail();
    string getPhone();
    void setName(string n);
    void setEmail(string e);
    void setPhone(string p);
};

class VehicleOwner : public User {
public:
    VehicleOwner();
    VehicleOwner(string n, string e, string p);
};

class Vehicle {
private:
    string plateNumber;
    string vehicleType;
    int year;
    VehicleOwner owner;
public:
    Vehicle();
    Vehicle(string plate, string type, int yr, VehicleOwner o);
    string getPlate();
    string getType();
    int getYear();
    VehicleOwner getOwner();
};

bool isValidEmail(string email);
bool isValidPhone(string phone);
bool isValidPlate(string plate);
```

```
void saveRecord(Vehicle v);  
vector<Vehicle> loadRecords();  
  
void addRecord();  
void displayAll();  
void searchRecord();  
  
#endif
```